Spotify YouTube

SPOTIFY & YOUTUPE MUSIC TRENDS

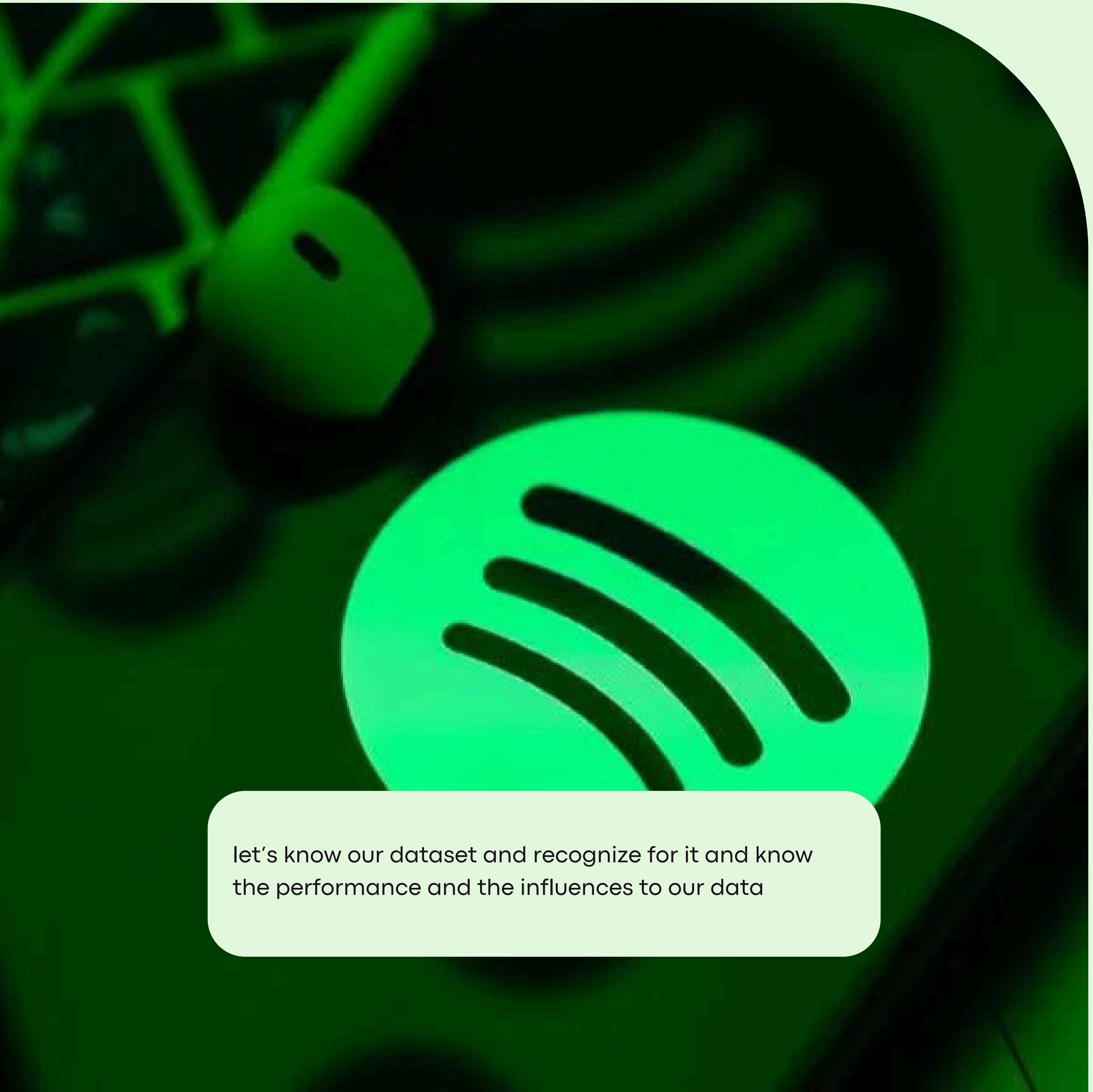
A complete end-to-end machine learning project that classifies songs and groups artist popularity into categories such as Low, Medium, High, and Super, using data preprocessing, feature engineering, and classification models in Python notebooks.

Presented By Abdelmoneim Adel

AGENDA

Data Overview

20718 ROWS



let's know our dataset and recognize for it and know the performance and the influences to our data

Background of the Study

Key columns:

- **Audio Features:** Danceability, Energy, Speechiness, Acousticness, etc.

- **Performance Metrics:** Spotify Streams; YouTube Views, Likes, Comments

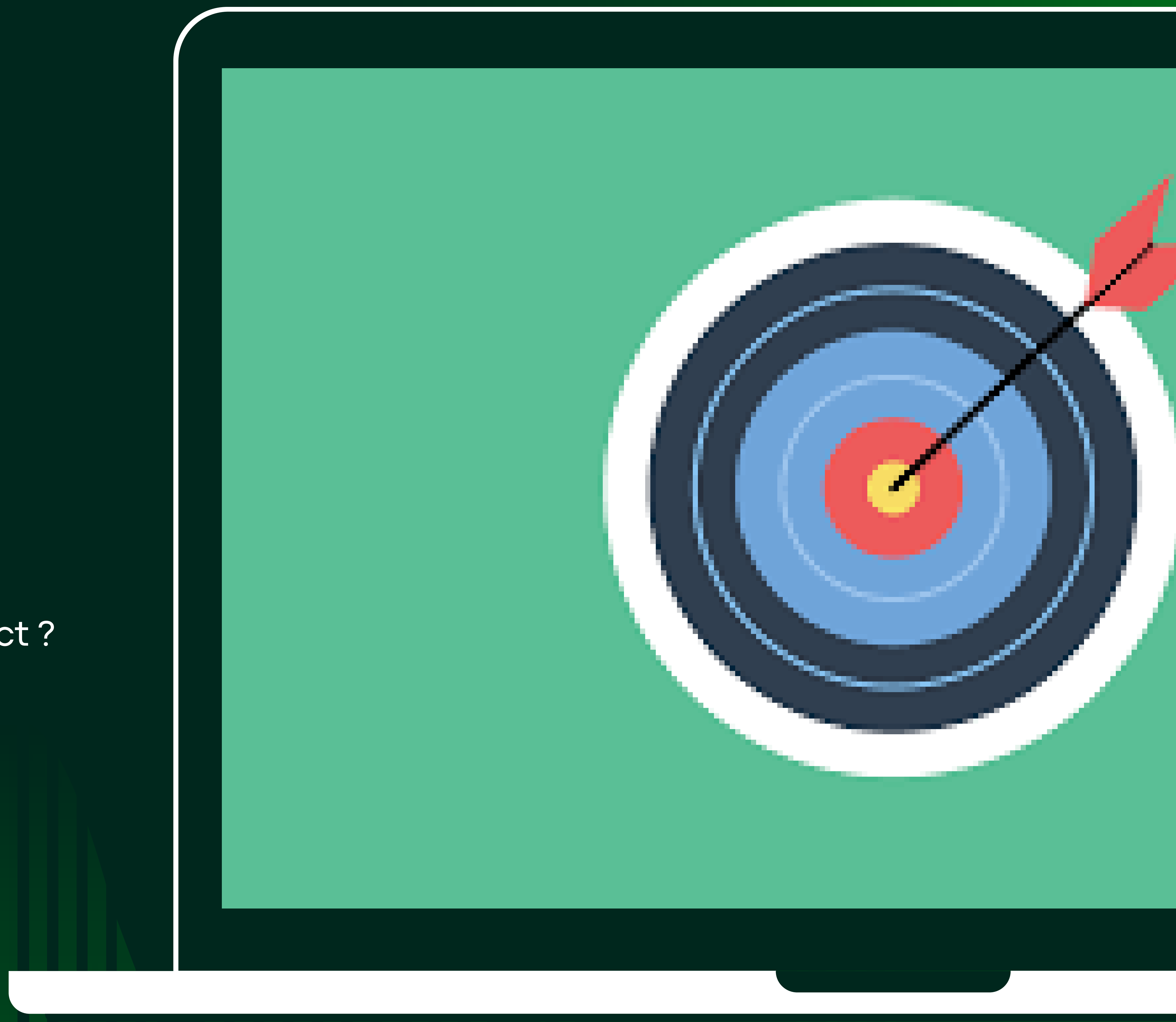
- **Metadata:** Track, Artist, Album, Album Type, Official Video flag, etc.

- **Source:** [Kaggle – Spotify and YouTube dataset].

Back to Agenda

Project Goals

Q1	Which platform is more popular ?
Q2	Top famous Artist and Albums ?
Q3	What is the most heard songs ?
Q4	Does the Danceability and Energy affect ?
Q5	Identify trends by artist, album type.



Project Pipeline

step by step from zero to hero

FIRST STEP

Data Understanding and Cleaning

SECOND SREP

Exploratory Data Analysis (EDA) and Visualization

THIRD STEP

Scaling and normalizing the data to start our training for models

Learning for models is simple if your model was clever and smart enough. by choosing the correct model and enough data to learn on.

Back to Agenda

Spotify & Youtube

Back to Agenda

FEATURE SELECTION PART

crucial step in the machine learning workflow, involving the process of choosing a subset of the most relevant features (input variables) from a dataset to be used in building a predictive model.

- Improve Model Performance
- Reduce Overfitting

RELEVANCE

How relevant is the feature to the target variable

REDUNDANCY

How much information does this feature provide that is not already provided by other features

Model Comparison

Top models which get the highest accuracies

LOGISTICREGRESSION	SVC	XGBCLASSIFIER
Accuracy: 0.912%	Accuracy: 0.989%	Accuracy: 0.997%

Back to Agenda

Classification Metrics

Report

Overall Classification Metrics:

	Model	Accuracy	Macro Avg Precision	Macro Avg Recall	Macro Avg F1-score
0	LogisticRegression	0.911829	0.912481	0.911817	0.911938
1	SVC	0.989409	0.989493	0.989390	0.989404
2	XGBClassifier	0.997662	0.997663	0.997657	0.997658

Project Key Insights

Artist

- Ed Sheeran knows how to make good music!
- Two out of the top 10 artists are Hispanic performers.
- CoComelon is a reference in children's music, worth exploring if you want to delve into this genre.

Songs

- "Despacito" is a worldwide case study as it has a melodic composition that plays interestingly with our brains. It is lively, simple, repetitive, and has a catchy rhythm.

Music is Haram in the Islamic Religion. And you shouldn’t hear it.
You are affected by what you listen to.

Make what satisfies our God, and your parents, then yourself.

For inquiries, contact us.

EMAIL abdelmoneim.adel5@gmail.com

SOCIAL MEDIA LinkTree/Abdelmoneim.Adel

CALL US 01556878109

Data Scientist